

Quiz 12

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Compute $\mathcal{L}\{t \cos 4t\}$ using the formula $\mathcal{L}\{tf(t)\} = -F'(s)$ where $F(s) = \mathcal{L}\{f(t)\}$.

$$\begin{aligned}
 \mathcal{L}\{t \cos 4t\} &= -\frac{d}{ds} [\mathcal{L}\{\cos 4t\}] \\
 &= -\frac{d}{ds} \left[\frac{s}{s^2 + 16} \right] \\
 &= -\frac{(s^2 + 16) - s(2s)}{(s^2 + 16)^2} \\
 &= \frac{s^2 - 16}{(s^2 + 16)^2}
 \end{aligned}$$

Problem 2. (5 points) Compute $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 9)}\right\}$ using the formula $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t)$ where $f(t) * g(t) = \int_0^t f(\tau)g(t - \tau) d\tau$, $\mathcal{L}^{-1}\{F(s)\} = f(t)$ and $\mathcal{L}^{-1}\{G(s)\} = g(t)$.

$$\begin{aligned}
 \mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 9)}\right\} &= \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} * \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 9}\right\} \\
 &= 1 * \frac{1}{3} \sin 3t \\
 &= \int_0^t \frac{1}{3} \sin 3(t - \tau) d\tau \\
 &= \frac{1}{9} \cos 3(t - \tau) \Big|_{\tau=0}^{\tau=t} \\
 &= \frac{1}{9} (1 - \cos 3t)
 \end{aligned}$$